



PRACTICAL WORK "NUMERICAL METHODS 2"  
MASTER 2 COMPUPHYS

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Rate equations or MC approaches for modeling  
growth

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## Introduction

The study of particle aggregation and deposition is essential for understanding growth processes in physical systems. Rate equations (RE) and Monte Carlo (MC) simulations are commonly used to model these systems, analyzing phenomena like nucleation, diffusion, and aggregation.

We examine a 2D system of single particles (monomers) and aggregates. Monomers move on the surface (rate  $D_1$ ) and can be deposited (rate  $F_A$ ) or die (rate  $F_D$ ). The mean density of monomers  $n_1$  changes over time according to:

$$\frac{dn_1}{dt} = F_A - F_D - 2D_1n_1^2 - D_1n_1n - 2\alpha F_A n_1 - \alpha F_A n \quad (0.1)$$

The first two terms represent monomer creation and destruction, the third and fourth terms describe the loss due to nucleation or aggregation, and the last two terms correspond to direct impingement.

The evolution of island density follows:

$$\frac{dn}{dt} = D_1n_1^2 + \alpha F_A n_1 \quad (0.2)$$

Rate equations like these are used in:

- **Surface Diffusion in Thin Film Growth:** Modeling nucleation and growth during physical vapor deposition (PVD).
- **Crystal Growth:** Studying epitaxial growth where atoms form ordered structures.
- **Biological Aggregation:** Modeling processes like protein cluster formation and bacterial colony growth.

## 1 Task 1: Resolution of the Rate Equations

We numerically solved the rate equations using the **Runge-Kutta method**, exploring the parameter  $F_A/D_1$ , which determines the competition between monomer deposition and diffusion-induced aggregation.

We will conclude

- Higher  $F_A/D_1$  favors rapid deposition and faster aggregation.
- Monomer death ( $F_D$ ) and impingement ( $\alpha$ ) reduce monomer availability, stabilizing island density over time.

Two cases were studied

### 1.1 Case 1: $F_A = 1$ , $F_D = 0$ , $\alpha = 0$

In this case:

- $F_A = 1$ : Constant deposition rate.
- $F_D = 0$ : No monomer death.
- $\alpha = 0$ : No direct impingement.

Figure 1 shows the time evolution of  $n_1$  (monomer density) and  $n$  (island density) for varying  $F_A/D_1$  ratios.

We observe the following:

- $n_1$  increases rapidly at first, then stabilizes at lower values, trending toward zero as  $F_A/D_1$  increases.
- $n$  grows indefinitely, with higher  $F_A/D_1$  accelerating island formation.

From Figure 1, we observe:

- **Rapid Increase of  $n_1$ :** Initially, monomers dominate since they are being deposited, and aggregation has not yet significantly reduced their density.
- **Stabilization of  $n_1$ :** As aggregation progresses,  $n_1$  decreases due to its conversion into islands. Higher  $F_A/D_1$  means more monomers are deposited relative to their ability to diffuse and aggregate, leading to a

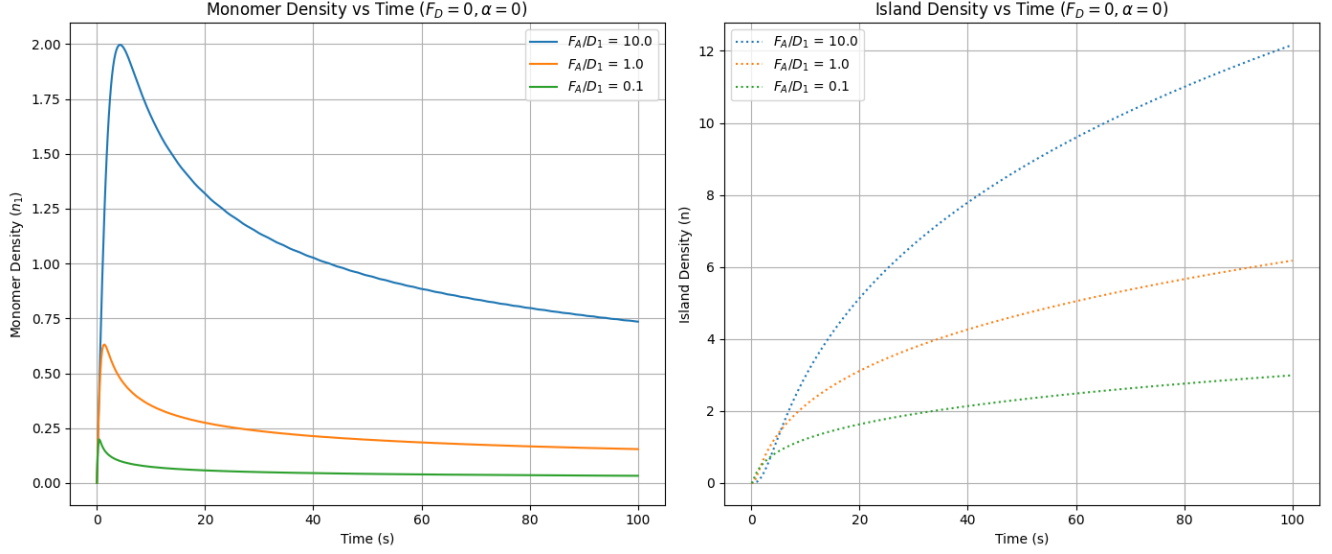


Figure 1: Time evolution of  $n_1$  and  $n$  for different  $F_A/D_1$  values ( $F_D = 0$ ,  $\alpha = 0$ ).

slower decline in  $n_1$ .

- **Growth of  $n$ :** Island density grows indefinitely because  $F_D = 0$ , so there is no mechanism to remove islands, and aggregation is fueled by a constant influx of monomers.

## 1.2 Case 2: $F_A = 1$ , $F_D = 0.5$ , $\alpha = 0.3$

In this case:

- $F_A = 1$ : Constant deposition rate.
- $F_D = 0.5$ : Monomer death rate.
- $\alpha = 0.3$ : Direct impingement parameter.

Figure 2 shows the time evolution of  $n_1$  (monomer density) and  $n$  (island density) for varying  $F_A/D_1$  ratios.

From Figure 2, we can observe that

- **Impact of  $\alpha$ :** The term  $2\alpha F_A n_1$  drives down  $n_1$  by increasing the rate of direct monomer-to-island aggregation. This effect is pronounced, especially with higher  $\alpha$ , and accelerates the reduction of  $n_1$ .
- **Stabilization of  $n$ :** With  $F_D > 0$ , island density  $n$  stabilizes because islands can now "die" or be removed. The interplay of  $F_A$ ,  $D_1$ , and  $F_D$  sets an equilibrium density for  $n$ .

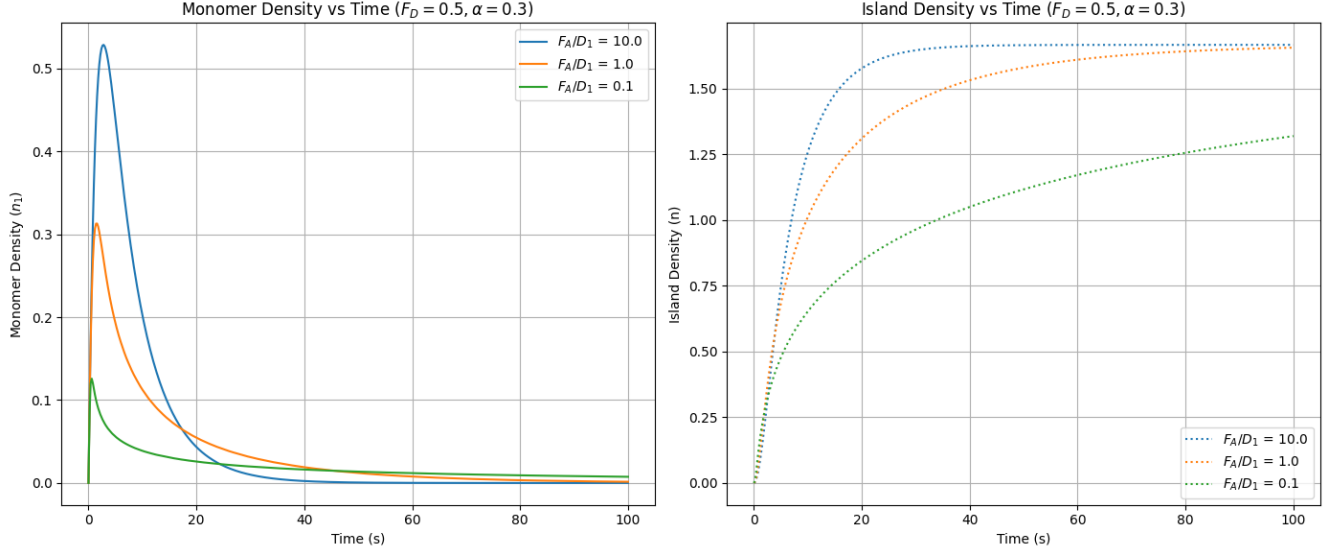


Figure 2: Time evolution of  $n_1$  and  $n$  for different  $F_A/D_1$  values ( $F_D = 0.5, \alpha = 0.3$ ).

## 2 Task 2: Anisotropy of Diffusion and Deposition

In this task, we explored the anisotropic effects of deposition and diffusion on monomer and island densities on a 2D grid. The simulation considered three scenarios: isotropic deposition and diffusion, localized deposition, and anisotropic diffusion.

### Simulation Setup

The system consists of a  $100 \times 100$  grid with periodic boundary conditions. Random deposition was modeled at a constant rate  $N_{diff} = 100$ , and monomers were allowed to diffuse between neighboring grid sites. Aggregation occurs when two monomers occupy adjacent sites or when a monomer collides with an existing island. For localized deposition, monomers were deposited in a restricted central region of size  $l \times l$ , where  $l = L/10$ . Anisotropic diffusion was modeled by allowing different rates of diffusion along the  $x$ - and  $y$ -axes ( $N_{diff}(x) = 1000, N_{diff}(y) = 10$ ).

### Results and Observations

Figure 3 shows the evolution of monomer and island densities over time, as well as the final island configurations for each case.

#### 1. Isotropic Deposition and Diffusion:

- Monomer density remained low throughout the simulation, as monomers were quickly consumed to form islands.
- Island density increased steadily, reflecting continuous aggregation.
- The final island distribution was uniform across the grid, as shown in the isotropic plot.

#### 2. Localized Deposition:

- Monomer density initially increased but stabilized as aggregation dominated.
- Island density grew significantly within the localized region, leading to dense clustering of islands.
- The final configuration demonstrates highly localized island growth, limited to the deposition area.

#### 3. Anisotropic Diffusion:

- Monomer density remained low, similar to the isotropic case, but island growth was faster in one direction

due to anisotropic diffusion rates.

- Islands exhibited elongated shapes, reflecting preferential diffusion along the  $x$ -axis.
- The final configuration displays distinct anisotropic island morphology, as shown in the anisotropic plot.

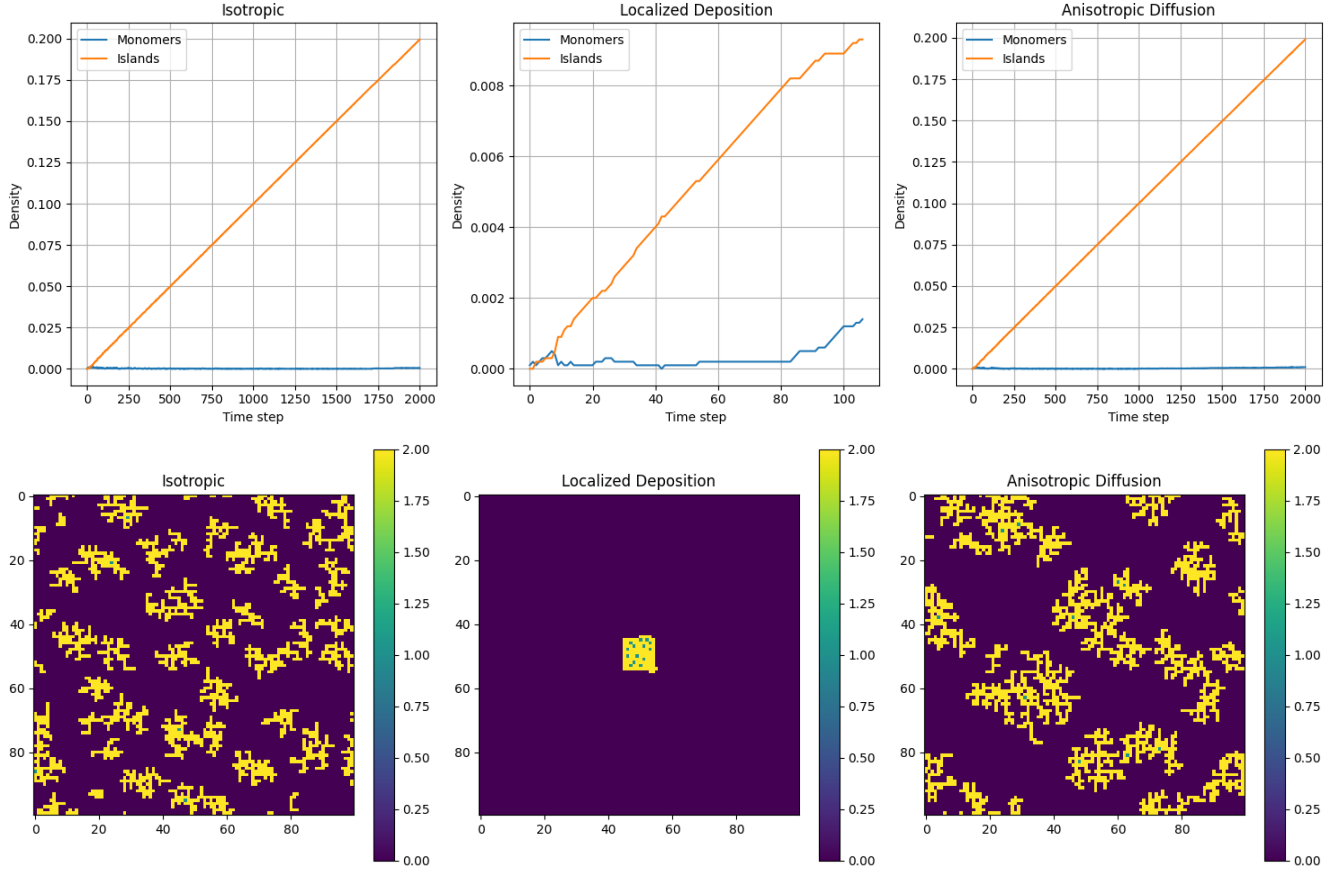


Figure 3: Evolution of monomer and island densities (top row) and final island configurations (bottom row) for isotropic deposition and diffusion, localized deposition, and anisotropic diffusion.

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### 3 Task 3: Analysis of Mixing Dynamics

The goal of this task was to simulate the diffusion dynamics of two species, A and B, and analyze the effect of the diffusion coefficient ratio  $D_B/D_A$  on the mixing index. The mixing index serves as a quantitative measure of the degree of mixing between the two species over time.

#### Methodology

The diffusion process was modeled using a grid-based approach, where species A and B diffuse with respective diffusion coefficients  $D_A$  and  $D_B$ . The simulations were performed for a range of  $D_B/D_A$  values, spanning orders of magnitude from  $10^{-2}$  to  $10^2$ .

To improve computational efficiency, the simulation code was optimized using the `Numba` library, enabling JIT (Just-In-Time) compilation. Each simulation was initialized with a random distribution of species on a  $100 \times 100$  grid. The system was evolved for 1000 timesteps for each value of  $D_B/D_A$ . The mixing index was computed at each timestep as follows:

$$\text{Mixing Index} = \sqrt{\frac{1}{L^2} \sum_{i,j} (c_A(i,j) - \langle c_A \rangle)^2}, \quad (3.1)$$

where  $c_A(i,j)$  represents the local concentration of species A at grid cell  $(i,j)$ ,  $\langle c_A \rangle$  is the mean concentration of A across the grid, and  $L$  is the grid size.

The average mixing index for each simulation was calculated by averaging the mixing indices over all timesteps.

#### Results

The plot in Figure 4 illustrates the relationship between the average mixing index and the diffusion coefficient ratio  $D_B/D_A$ . The results show variability in the mixing index, with noticeable peaks around certain values of  $D_B/D_A$ . The key observations include:

- Minimal mixing at extreme ratios ( $D_B/D_A \ll 1$  or  $D_B/D_A \gg 1$ ), likely due to the dominance of one species' diffusion over the other.
- Maximum mixing index observed near  $D_B/D_A \approx 1$ , indicating optimal conditions for inter-species diffusion.

In addition to the mixing index plot, the final grid configurations for different  $D_B/D_A$  values were recorded. These configurations, shown in Figure 5, illustrate the spatial distribution of species A and B at the end of the simulation.

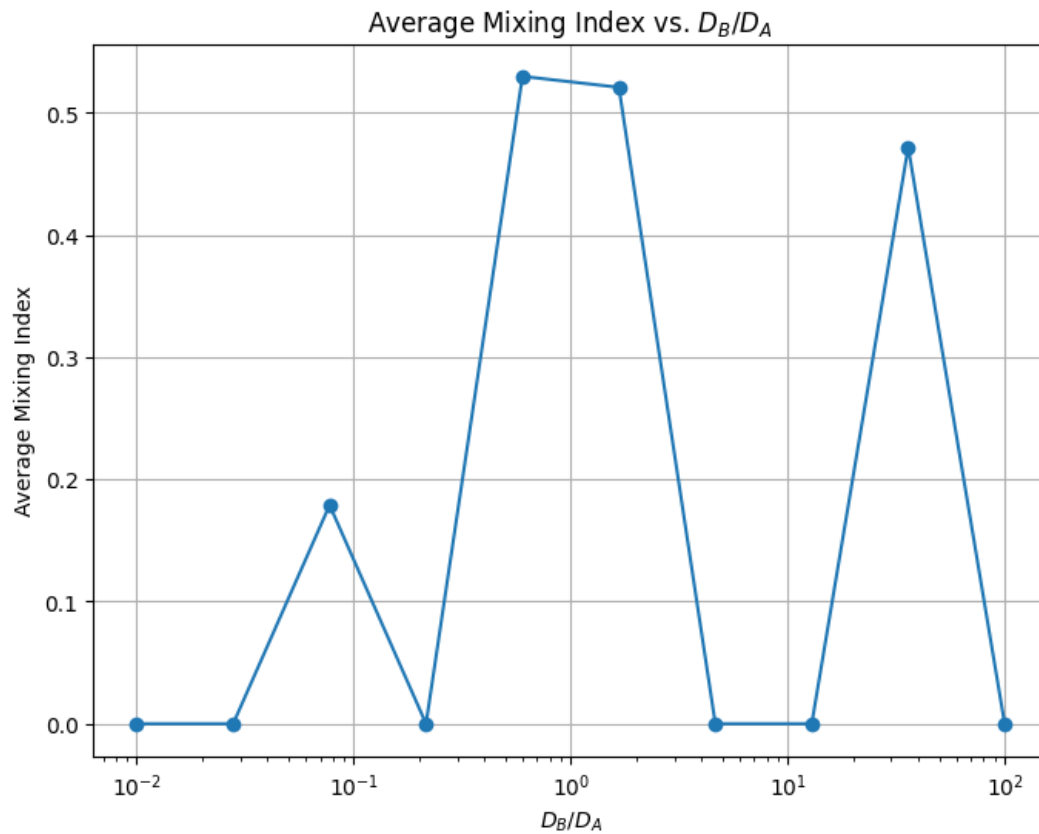


Figure 4: Average Mixing Index as a function of  $D_B/D_A$ .

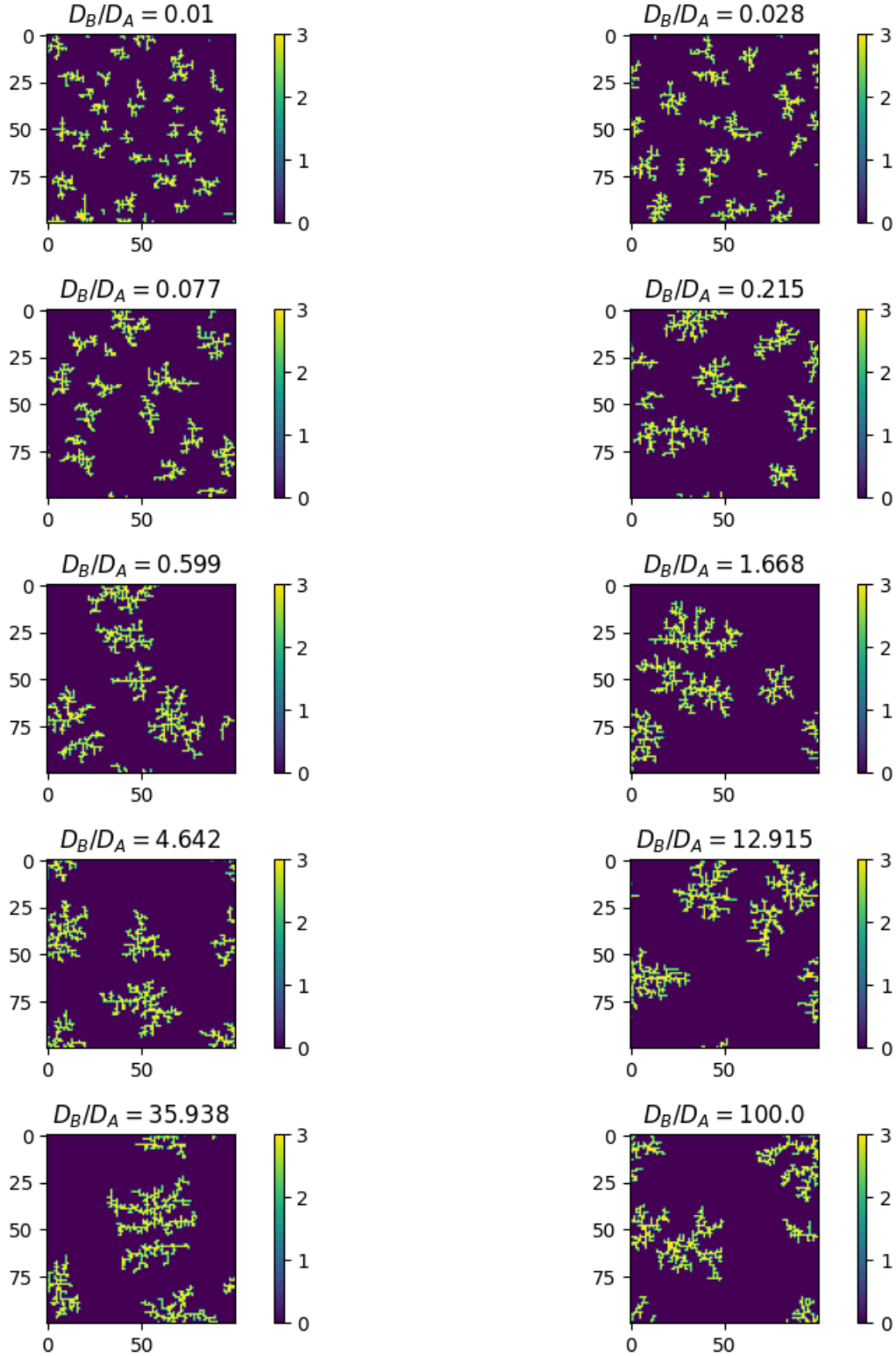


Figure 5: Final grid configurations for different  $D_B/D_A$  ratios. Each subfigure corresponds to a specific ratio, with the concentrations of species  $A$  and  $B$  visualized.

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## 4 Task 4: Species that can undergo chemical reaction

In this task, we investigated the behavior of two species,  $A$  and  $B$ , which react upon contact to form a new species  $C$  on a  $L \times L$  surface grid. The objective was to explore the influence of the diffusion coefficient ratio  $D_B/D_A$  on the formation of  $C$  molecules, considering deposition, diffusion, and reaction dynamics. To ensure small coverage and avoid double-layer growth, the surface coverage was kept below 0.2.

### Model Description

The simulation was implemented as follows:

- Molecules  $A$  and  $B$  were deposited at random positions on the surface with a flux  $F = 1$  layer/s.
- The molecules  $A$ ,  $B$ , and  $C$  diffused to neighboring lattice sites with coefficients  $D_A$ ,  $D_B$ , and  $D_C$ , respectively. The diffusion coefficients were scaled to the simulation time step  $\Delta t$ .
- When an  $A$  and  $B$  molecule occupied adjacent lattice sites, they irreversibly reacted to form  $C$ , which was placed at the average position of the reactants.
- Species  $C$  diffused at a fixed rate of  $D_C = 10 \text{ s}^{-1}$ .

### Simulation Parameters

The parameters used in the simulation are summarized in Table 1.

| Parameter                              | Value                            |
|----------------------------------------|----------------------------------|
| Grid size ( $L \times L$ )             | $100 \times 100$                 |
| Deposition flux ( $F$ )                | 1 layer/s                        |
| Diffusion coefficient of $A$ ( $D_A$ ) | $10^4 \text{ s}^{-1}$            |
| Diffusion coefficient of $B$ ( $D_B$ ) | Varied ( $0.01D_A$ to $100D_A$ ) |
| Diffusion coefficient of $C$ ( $D_C$ ) | $10 \text{ s}^{-1}$              |
| Time step ( $\Delta t$ )               | $1 \times 10^{-4} \text{ s}$     |
| Total simulation time                  | 10 s                             |
| Number of independent runs             | 5                                |

Table 1: Simulation parameters for Task 4.

### Results and Discussion

To study the effect of the diffusion coefficient ratio  $D_B/D_A$  on the formation of  $C$  molecules, simulations were conducted for values of  $D_B/D_A$  logarithmically spaced between  $10^{-2}$  and  $10^2$ . For each ratio, the final count of  $C$  molecules was averaged over 5 independent runs to account for stochastic fluctuations.

Figure 6 shows the relationship between  $D_B/D_A$  and the mean number of  $C$  molecules formed on the surface. The results reveal a non-linear dependence of  $C$  formation on the diffusion ratio:

- For very low  $D_B/D_A$  ratios ( $D_B \ll D_A$ ),  $B$  molecules diffused too slowly relative to  $A$ , limiting the reaction rate and resulting in fewer  $C$  molecules.
- At intermediate ratios ( $D_B/D_A \approx 1$ ), the diffusion rates of  $A$  and  $B$  were balanced, facilitating optimal interaction and leading to a peak in  $C$  formation.
- For high  $D_B/D_A$  ratios ( $D_B \gg D_A$ ),  $B$  molecules diffused too rapidly, reducing the likelihood of encounters with  $A$ , which again limited the formation of  $C$ .

The simulation highlights the importance of balancing the diffusion rates of reactive species to maximize product formation. Peaks observed near  $D_B/D_A \approx 5 - 10$  indicate an optimal regime where diffusion rates enable frequent encounters between  $A$  and  $B$ . Furthermore, the variability in  $C$  counts, captured by error bars (not shown here

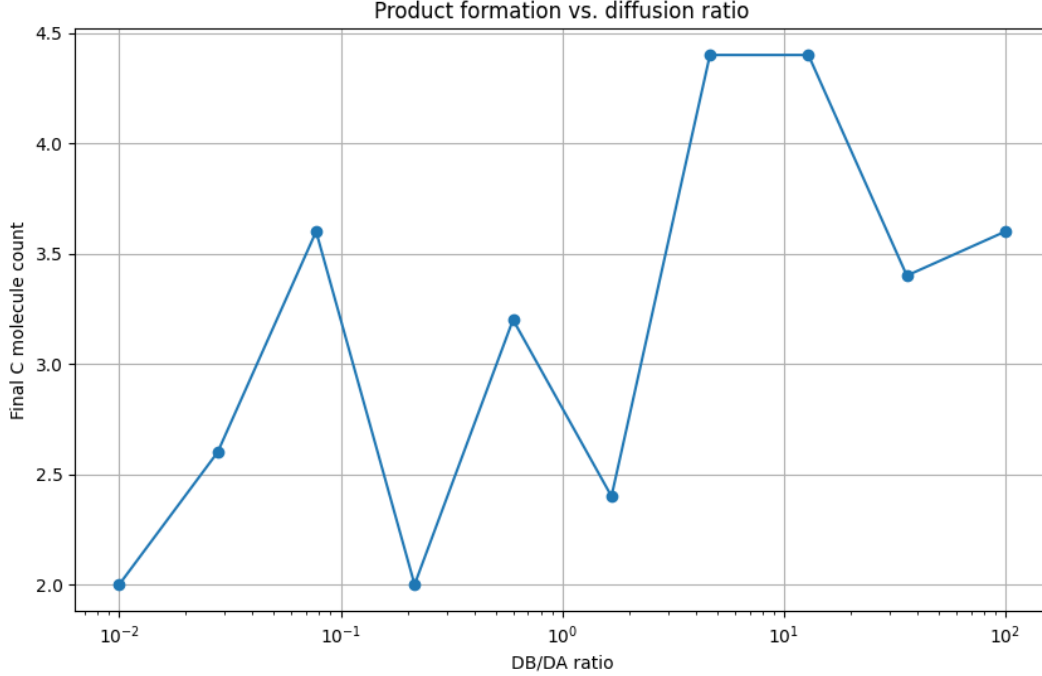


Figure 6: Dependence of the final count of  $C$  molecules on the diffusion coefficient ratio  $D_B/D_A$ .

for simplicity), suggests that stochastic effects play a significant role in the reaction dynamics, particularly for low  $D_B/D_A$  values.

## 5 Conclusion

This study provided an in-depth exploration of particle aggregation, deposition, diffusion, and reaction dynamics in 2D systems. We investigated various physical scenarios, yielding the following key insights:

- **Task 1: Rate Equations**

Solving the rate equations revealed the competing roles of deposition, diffusion, and aggregation in determining the time evolution of monomer and island densities. The results highlighted how higher deposition-to-diffusion ratios ( $F_A/D_1$ ) promote rapid aggregation, while monomer death ( $F_D$ ) and direct impingement ( $\alpha$ ) stabilize island density by limiting monomer availability.

- **Task 2: Anisotropic Deposition and Diffusion**

Simulations demonstrated that isotropic conditions lead to uniform island distribution, while localized deposition creates dense clusters. Anisotropic diffusion produced elongated island morphologies, reflecting directional preferences.

- **Task 3: Mixing Dynamics**

Analysis of the diffusion-driven mixing of two species revealed the importance of the diffusion coefficient ratio ( $D_B/D_A$ ) in optimizing mixing. The mixing index peaked near  $D_B/D_A \approx 1$ , where both species diffuse at similar rates, promoting effective inter-species interactions. Minimal mixing occurred at extreme ratios due to imbalances in diffusion rates.

- **Task 4: Chemical Reaction Dynamics**

Investigating reactive species on a lattice highlighted the non-linear dependence of product formation on the diffusion coefficient ratio ( $D_B/D_A$ ). Optimal product formation occurred at  $D_B/D_A \approx 10$ , where balanced diffusion rates maximized encounters between reactive species. Excessively low or high ratios reduced reaction efficiency, illustrating the need to balance diffusion rates for enhanced reactivity.